

GeoNews — Week 34/2026

Weekly digest on Quaternary palaeoclimate research, East African rift dynamics, palaeohydrology and time-series analysis · 17 August 2026

What is GeoNews? An **automatically generated weekly digest**. A research workflow scans the publications of the past one to two weeks along with funding, policy and community sources, scores the hits against a fixed set of criteria, and assembles this issue from them.

The thematic profile is **aligned with the research interests of Markus L. Fischer** (Palaeoclimate Dynamics group, University of Potsdam): Quaternary palaeoclimate and palaeoenvironmental reconstruction, the East African Rift, the African Humid Period and human-environment coupling, lake-level and water-balance modelling, palaeofloods and fluvial geomorphology, ICDP drill cores, and linear and non-linear time-series analysis. Anyone working in these fields will probably find something useful here — it is explicitly not a complete survey of the Quaternary sciences.

Selection, scoring and short texts are produced by machine and reviewed before publication. Details may contain errors; in case of doubt the linked original source prevails.

Each paper receives a score from 0–100 across five criteria: thematic proximity (40), methodological proximity (25), impact/venue (15), connectivity (10) and recency (10). Categories: ● from 75 must-read · ● 50–74 relevant · ● 30–49 radar. The score measures proximity to this newsletter's profile and prioritises reading — it is not a quality judgement on the individual paper.

In brief

Six scored papers in the ranking, plus two on the radar. At the top this week sits a clear ● must-read that is also the thematic bullseye: a **Holocene hydroclimate reconstruction of Lake Sidi Ali in the Moroccan Middle Atlas** with Redfit and wavelet time-series analysis (*Climate of the Past*). North African desert margin, a lacustrine archive, water-balance proxies and a spectral decomposition of the variability into separate forcing regimes — this hits several profile anchors at once and transfers directly to East African rift lakes and the frequency analysis applied there.

Behind it come five ● papers: a multi-proxy speleothem record from south-eastern Iberia on the **termination of the last North African Humid Period** (preprint), a Nature synthesis of **Holocene fire regimes in North America** from 140 charcoal records, a modelling study of the effect of **glacial Northern Hemisphere ice sheets on atmospheric circulation**, an **NAO reconstruction of the last millennium** from water-isotope archives, and a dendroclimatological **cold-season temperature series for central Japan** (preprint). On the radar, two *Climate of the Past* papers at the edge of the profile.

On how this issue came about — constrained this week. Crossref access from the compute environment was blocked as in previous weeks; in addition, the browser-based fetch interface was unavailable this time. The usual systematic ISSN query of the core journals via Crossref was therefore not possible. Instead, the review ran via web search plus the freely accessible publisher and journal landing pages (Copernicus, Nature open access, MDPI). This shifts coverage toward

open-access titles whose abstracts are directly readable; Elsevier and Wiley titles could not be reliably captured with title and created date without Crossref and are therefore **not** carried as a separate title-only block (rather than inventing results). The dates given are journal publication dates, not Crossref created dates. Every statement about content in the scored entries rests on a consulted abstract.

On funding, the focus is unchanged: the **ERC Starting Grant 2027** (deadline **14 Oct 2026**), the **PAGES applications (11 Sep 2026**, in just over three weeks) and the Horizon palaeoclimate call **HORIZON-CL5-2027-01-D1-08** (opens 17 Nov 2026, deadline 4 Mar 2027). In science policy, the **WissZeitVG amendment** adopted by the federal cabinet on 29 Jul 2026 remains the dominant topic; the Bundestag debate is expected after the summer recess in September.

Part A — Papers of the week

Captured were papers with a publication date from 4 August 2026 onward. DOIs already listed in earlier issues (through 12 Aug) are excluded. All scored entries rely on a consulted abstract. A block of Elsevier titles without abstract is omitted this week because the Crossref query needed for it was unavailable (see methods).

● **Must-read (from 75)**

1. Holocene hydroclimatic variability and multi-frequency analyses at Lake Sidi Ali (Morocco) · *Climate of the Past*, 2026 · **Score 93** A Holocene sediment record from Lake Sidi Ali in the semi-arid to sub-humid Middle Atlas — one of the most climate-sensitive stretches of the North African desert margin. The chronology rests on $^{210}\text{Pb}/^{137}\text{Cs}$ and a ^{14}C age on pollen concentrates; a high-resolution core-scanning XRF profile separates lake-internal (chemical precipitation) from lake-external (detrital input) processes. Redfit and wavelet analyses reveal distinct millennial- to centennial-scale periodicities, grouped via correlation analysis of highly significant spectra into three "Redfit Proxy Groups" (RPGs). Result: two partly decoupled regimes — RPG 1 (Ca, Sr, Ca/Ti, Sr/Ti) with 2- and 1-kyr periodicities, interpreted as a precipitation/evaporation signal in the context of North Atlantic and solar forcing, and RPG 2 (Fe, Ti, K, Si/Ti) with 3.5- and 1.5-kyr periodicities, interpreted via lake productivity and detrital input. Connective as a core Tier-1 theme (Holocene hydroclimate and water-balance reconstruction at the desert margin) and methodologically as a clean example of spectral time-series decomposition on a lacustrine multiproxy archive — the same Redfit/wavelet toolkit and the same question of separable forcing regimes arise directly at East African rift lakes. *Theme 40* · *Method 22* · *Venue 11* · *Connectivity 10* · *Recency 10* (publ. 10 Aug 2026) · DOI: [10.5194/cp-22-1537-2026](https://doi.org/10.5194/cp-22-1537-2026)

● **Relevant (50-74)**

2. Early to Middle Holocene aridification trend in southeastern Iberia and the termination of the last North African Humid Period revealed by a multi-proxy speleothem record · *EGUsphere* (preprint, under review for *Climate of the Past*), 2026 · **Score 73** A multi-proxy stalagmite from Simarrón II cave (south-eastern Iberia), 9.7–4.4 ka BP, combining calcite $\delta^{18}\text{O}/\delta^{13}\text{C}$, Mg/Ca, petrography, fluid-inclusion isotopes ($\delta^{18}\text{O}_{\text{fi}}$, $\delta^2\text{H}$) and TEX_{86} palaeotemperatures. Result: between 9.7 and 6.0 ka, low $\delta^{18}\text{O}_{\text{carb}}$, $\delta^{13}\text{C}$ and Mg/Ca values indicate relatively wet conditions; from ~6.0 ka the proxies rise progressively and stabilise after ~5.5 ka — a shift to arid conditions **within about 500 years**, which the authors link to the termination of the last North African Humid Period. TEX_{86} temperatures (13–17 °C) exceed the

present-day cave value (11.7 °C). Connective as a Tier-1 theme (AHP termination and its pace): the same methodological question about the tempo and spatial coherence of the wet-to-dry transition that is central in East African archives — here through a European speleothem rather than an African lake sediment, and therefore with thematic rather than regional connectivity. Preprint, hence the low venue weight; the finding stands provisionally pending completion of review. *Theme 40 · Method 17 · Venue 4 · Connectivity 6 · Recency 6 (publ. 7 Aug 2026) · DOI: 10.5194/egusphere-2026-4435*

3. Human climatic and ecological drivers of Holocene fire regimes in North America · *Communications Earth & Environment*, 2026 · **Score 69** A synthesis of 140 Holocene-length (12 kyr) charcoal records via a systematic database search with subsequent sensitivity analysis; ecoregion-scale spatiotemporal variations are compared with reconstructed climate parameters, global CO₂ emissions, ecological productivity and human landscape impacts, complemented by a taphonomically corrected comparison with archaeological data. Result: temperature, emissions, vegetation and human impacts each drove Holocene fire activity; the authors date the onset of human influence on the fire regime to **about 6,500 cal yr BP** — earlier than previous estimates. Connective as a multiproxy palaeoecology study on human–environment coupling: separating climatic from anthropogenic signals in a noisy proxy (charcoal) across many sites is methodologically related to the attribution question in continental palaeoenvironmental archives. The high venue weight (Nature family) carries the score at only moderate thematic proximity. *Theme 25 · Method 15 · Venue 13 · Connectivity 6 · Recency 10 (publ. 12 Aug 2026) · DOI: 10.1038/s43247-026-03918-w*

4. The influence of glacial Northern Hemisphere ice sheets on atmospheric circulation · *Climate of the Past*, 2026 · **Score 64** A modelling study of the effect of the glacial Northern Hemisphere ice sheets on global atmospheric circulation. Result: the ice sheets shifted the North Atlantic winds southward and increased summer rainfall across Eurasia; the height and shape of the ice sheets had a much greater influence on global wind and rainfall patterns than changes in Earth's orbit, greenhouse gases or surface albedo. Connective as a Tier-2 climate-dynamics study: the question of which boundary condition (ice-mass topography vs orbital/radiative forcing) to attribute the rainfall-relevant circulation change to touches the forcing decomposition that is also central to glacial African monsoon and water-balance dynamics. *Theme 25 · Method 16 · Venue 11 · Connectivity 6 · Recency 6 (publ. 5 Aug 2026) · DOI: 10.5194/cp-22-1457-2026*

5. Multidecadal behavior of the North Atlantic Oscillation during the last millennium · *Climate of the Past*, 2026 · **Score 63** A new reconstruction of the North Atlantic Oscillation (NAO) over the last millennium from water-isotope proxy data from ice cores, speleothems, tree rings and other archives. Result: pronounced but irregular multidecadal to centennial variability; the authors stress that better representation of this low-frequency variability in climate models is necessary for robust projections. Connective as a Tier-2 study reconstructing a mode of variability from a multi-archive isotope network — methodologically related to the question of how robust low-frequency signals are to archive and calibration choices, which also holds for continental hydroclimate series. *Theme 25 · Method 15 · Venue 11 · Connectivity 6 · Recency 6 (publ. 4 Aug 2026) · DOI: 10.5194/cp-22-1423-2026*

6. A cold-season temperature reconstruction for central Japan since 259 CE from a two-millennia Kiso *Chamaecyparis* ring-width chronology · *EGUsphere* (preprint, under review for *Climate of the Past*), 2026 · **Score 52** A near-1,800-year reconstruction of November-to-April temperature in central Japan combining tree rings from living trees, archaeological wood, historic buildings and buried wood. The series most reliably captures interannual to decadal variability, identifies sustained warmth from 1999 to 2015, and remains more uncertain in longer-

term variations. Connective as a geochronological-dendroclimatological case study (chronology building from heterogeneous wood sources, calibration, uncertainty estimation) at the edge of the profile; preprint, hence the low venue weight. *Theme 25 · Method 14 · Venue 4 · Connectivity 3 · Recency 6 (publ. 7 Aug 2026) · DOI: [10.5194/egusphere-2026-4506](https://doi.org/10.5194/egusphere-2026-4506)*

● Radar

- **Amplified cooling of Snowball Earth from a salt-albedo feedback** · *Climate of the Past* · **44** ● · Implementation of a salt-albedo feedback in a simple climate model; the feedback could have amplified cooling during the initial phase of a Snowball Earth state. Noted as a deep-time climate-dynamics study (feedback and tipping-point mechanics) at the edge of the profile. DOI: [10.5194/cp-22-1499-2026](https://doi.org/10.5194/cp-22-1499-2026)
- **Atmospheric pressure and anemological conditions in south-western Greenland in the second half of the 18th century** · *Climate of the Past* · **42** ● · Analysis of some of the oldest meteorological series in the region (pressure, wind conditions) and comparison with present-day measurements. Noted as a historical-climatology study (early-instrumental data rescue) at the edge of the profile. DOI: [10.5194/cp-22-1441-2026](https://doi.org/10.5194/cp-22-1441-2026)

Part B — Funding & calls

ERC Starting Grant 2027 — deadline 14 Oct 2026. The call (ERC-2027-STG) has been open since **22 Jul 2026**. Eligibility: PhD defence between 1 Jan 2017 and 31 Dec 2026 (up to ten years before the reference year), extendable for parental and care leave. Funding up to EUR 1.5 million over five years (plus up to EUR 1 million for start-up/equipment). Step-1 results in April 2027, interviews and Step-2 results in August 2027. New from the 2027 calls: at most one Starting and one Consolidator Grant per career. The **ERC Consolidator 2027** is expected for January 2027. Sources: [ERC — Starting Grant](#) · [ERC — Applying in the 2027 competitions](#).

PAGES — working-group and workshop support, deadline 11 Sep 2026. Next application round for workshops, meetings, new working groups and extensions of existing groups; applicants should contact an SSC member at least two weeks before the deadline, and workshop support should be requested at least six months before the event. Ahead of the Horizon palaeoclimate call, a convenient instrument for forming a consortium visibly. In addition: **PAGES Early-Career Award 2027**, deadline **29 Sep 2026**. Sources: [PAGES — Workshop/meeting support](#) · [PAGES Newsletter July 2026](#).

Horizon Europe — "Palaeoclimate science for a better understanding of Earth system dynamics" (HORIZON-CL5-2027-01-D1-08). Opens **17 Nov 2026**, deadline **4 Mar 2027**, Research and Innovation Action, budget about **EUR 14 million for roughly two projects**, i.e. some EUR 7 million per consortium. Aim per the call text: robust information on palaeoclimate states and events outside the variability of recent centuries, as a basis for advancing Earth system models. With only two projects funded Europe-wide the competition is fierce; consortium talks should be started early. Sources: [Topic page](#) · [CL5 Work Programme 2026–2027 \(PDF\)](#).

ICDP — proposal cycle. The regular annual deadline for full, pre- and workshop proposals is **15 January**; the 2027 cycle is likely to follow the same pattern, though no formal announcement has appeared. Workshop calls run through the year. Source: [ICDP — Proposals](#).

DFG — changed framework conditions. No new palaeoclimate-specific call. Important for planning: the **programme allowance rises from 22 to 25 percent from 2027**; the next

Priority Programme application round will only take place at the **end of 2026**, with funding starting from **2028**. Additionally, at the margin of subject funding: applications for the **DFG Communicator Award 2027** (science communication) are open from **3 Aug to 30 Sep 2026**. Sources: [Forschung & Lehre — 25 instead of 22 percent from 2027](#) · [DFG — Press releases 2026](#).

Part C — Science policy (DE/EU)

WissZeitVG — amendment adopted by the cabinet on 29 Jul 2026, Bundestag to follow in the autumn. The federal cabinet adopted the bill to modernise fixed-term employment law in science on 29 July 2026; the May draft was revised in parts after criticism from associations. At its core are, for the first time, binding **minimum contract durations: three years for doctoral researchers' first contracts, two years for postdocs**; the maximum fixed-term duration in the postdoc phase is **standardised at six years**. For **student research assistants** the draft provides a minimum term of one year and raises the maximum fixed-term duration from six to eight years. According to parliamentary sources, the Bundestag debate is to be expected after the summer recess, i.e. in September. As interpretation, not established fact: minimum durations improve the predictability of early contracts, while the removal of earlier carry-over flexibility further narrows individual fixed-term planning in the postdoc phase — the further parliamentary path remains to be watched. Sources: [BMFTR — Cabinet adopts amendment to the WissZeitVG](#) · [Bundestag — Reform of the Wissenschaftszeitvertragsgesetz](#) · [Wiarda blog — WissZeitVG: BMFTR presents draft](#).

Federal budget 2027 — cabinet decision of 6 Jul 2026, Bundestag to follow in the autumn. The government draft has stood since 6 July 2026. The BMFTR core budget is about **EUR 22 billion** — a nominal increase of only about **0.7 percent** over 2026. Real increases are almost exclusively in the special fund for infrastructure and climate neutrality, whose R&D investments rise markedly in 2027 over 2026. The first parliamentary reading is due in the autumn. As interpretation: the regular budget stagnates in real terms given current pay and materials cost rises, keeping the operating situation of the funding organisations under pressure. Sources: [Forschung & Lehre — Only a slim increase for the BMFTR core budget](#) · [Table.Briefings — Budget 2027](#).

DFG cuts at the postdoc level remain in force. Owing to high pay rises and increased equipment and consumables costs that the promised annual funding increments do not cover, the DFG has postponed the funding start of new Research Training Group applications and **suspended the approval of postdoc positions**. For postdoc career paths in the geosciences this removes a funding route for now, increasing the pressure on individual-grant and EU funds. Source: [Fakultätentag Informatik](#).

Part D — Community & dates

- **PAGES working-group and workshop applications — 11 Sep 2026** (in just over three weeks) · **PAGES Early-Career Award 2027 — 29 Sep 2026**.
- **DEUQUA meeting 2026 — 28 Sep to 2 Oct 2026 in Freiburg im Breisgau**, theme "Environmental Records and Landscape Change", with full-day excursions on 28 Sep and 2 Oct (Black Forest/Vosges glaciation, Rhine Valley loess, fluvial dynamics in the Upper Rhine Graben, among others). Source: [DEUQUA 2026](#).

- **DFG Communicator Award 2027** — deadline **30 Sep 2026**.
 - **ERC Starting Grant 2027** — deadline **14 Oct 2026**.
 - **Horizon CL5 palaeoclimate call** — opens **17 Nov 2026**, deadline **4 Mar 2027**.
 - **AGU Fall Meeting 2026** — **7–11 Dec 2026 in San Francisco**; the abstract deadline (5 Aug 2026) has passed. Source: [AGU](#).
 - **EGU General Assembly 2027** — **4 to 9 Apr 2027 in Vienna**, hybrid. Abstract deadline not yet published; typically mid-January. Source: [EGU](#).
 - **ICDP** — next regular proposal deadline expected **15 Jan 2027**, following previous years and not yet confirmed.
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Methods & sources

The paper review follows the fixed set of criteria (thematic proximity, methodological proximity, impact/venue, connectivity, recency) and excludes DOIs already listed in an earlier issue. Funding opportunities, science policy and dates rely on the sources linked in Parts B to D.

On how this issue came about — a different research path. Crossref access from the compute environment was blocked as in previous weeks; in addition, the browser-based fetch interface otherwise used as a fallback was unavailable this week. The systematic ISSN query of the core journals via the Crossref REST API could therefore not run. The review instead used web search combined with the freely accessible "recent" lists and article landing pages of the open-access journals — in particular the Copernicus layer (*Climate of the Past*, EGU sphere preprints), Nature open access (*Communications Earth & Environment*) and MDPI. For every scored entry the abstract was consulted on the respective landing page.

What this means for coverage. The selection is shifted this week toward freely accessible titles because their abstracts are directly readable. The Elsevier (QSR, EPSL, GPC, P³, Boreas as Wiley, Catena, Geomorphology) and Wiley core journals could not be reliably captured with title and date within the window without Crossref; a title-only block without abstracts is therefore omitted rather than deriving results from incomplete hit lists. The dates given are **journal publication dates** (landing page), not Crossref `created` dates; the two usually differ by a few days. A direct East Africa / rift / AHP bullseye from an African primary study was not findable within the window; the profile is served through the North African desert margin (Sidi Ali, Simarrón II) and the methods side.

GeoNews is an automatically generated digest, published by Dr Markus L. Fischer, Palaeoclimate Dynamics group, University of Potsdam. The thematic profile follows his research interests; selection is made by a rule-based workflow, not an editorial team. The digest is an independent literature overview with no connection to the publishers or funding bodies named, and constitutes no recommendation or advice. Corrections and tips are welcome.